

* Revision: Row Picture & Column Picture

$$\begin{cases} x + 2y = 3 \\ 4x + 5y = 6 \end{cases}$$

← Systems of Linear equation in 2 variables

Row Picture

→ Two lines intersect at a Point

→ In general for m equations and n variables: It is the point of intersection of m $(n-1)$ d hyperplane in n dimensional space.

Column Picture

→ Linear combination of column vectors for

A in $Ax = b$

$$\begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

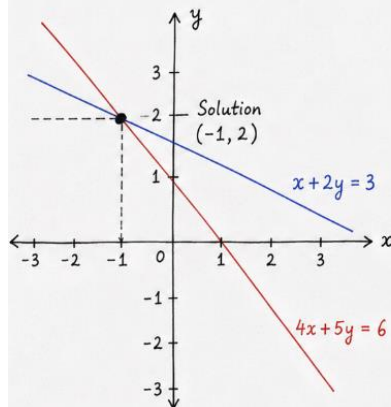
$$x \begin{bmatrix} 1 \\ 4 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

satisfies for

$$\begin{aligned} x &= -1 \\ y &= 2 \end{aligned}$$

1) ROW PICTURE (Lines / Equations)

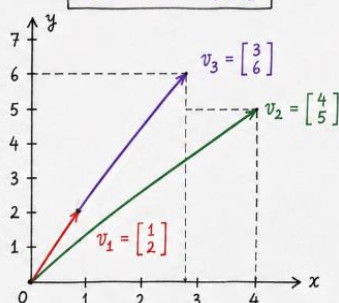
$$\begin{aligned} \text{Line 1: } x + 2y &= 3 \\ \text{Line 2: } 4x + 5y &= 6 \end{aligned}$$



Solution (Intersection)
 $x = -1, y = 2$

2) COLUMN PICTURE (Vectors / Columns)

$$\begin{aligned} v_1 &= \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ v_2 &= \begin{bmatrix} 4 \\ 5 \end{bmatrix} \\ v_3 (\text{target}) &= \begin{bmatrix} 3 \\ 6 \end{bmatrix} \end{aligned}$$



Target vector: $v_3 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$
(We want to reach v_3 using v_1 and v_2)

3) VECTOR ADDITION / COMBINATION

(Triangle / Parallelogram Law)

Find scalars a, b such that
 $av_1 + bv_2 = v_3$



Using the solution $(x, y) = (-1, 2)$:

$$\begin{aligned} -1 \cdot v_1 + 2 \cdot v_2 &= v_3 \\ -1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} 4 \\ 5 \end{bmatrix} &= \begin{bmatrix} -1 \\ -2 \end{bmatrix} + \begin{bmatrix} 8 \\ 10 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix} \end{aligned}$$

→ A system of Linear equation can be written in matrix form as $Ax = b$

$$\begin{bmatrix} \uparrow & \uparrow & \uparrow \\ a_1 & a_2 & a_3 \\ \downarrow & \downarrow & \downarrow \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

$$x_1 \begin{bmatrix} \uparrow \\ a_1 \\ \downarrow \end{bmatrix} + x_2 \begin{bmatrix} \uparrow \\ a_2 \\ \downarrow \end{bmatrix} + x_3 \begin{bmatrix} \uparrow \\ a_3 \\ \downarrow \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

↑ Linear Combination of Columns of A gives Column vector b.

→ Homogeneous system of linear equation

$$Ax = 0$$

Example: $2x + y - z = 0$
 $x - y + z = 0$

→ Non-Homogeneous system of linear equation

$$Ax = b \text{ and } b \neq 0$$

Example: $2x + y = 3$
 $x - y = 0$

→ The system of Linear equations $Ax = b$ can have

- ① Unique solution
- ② Infinitely many solution
- ③ No solution

→ The system of Linear equations $Ax = 0$ always has some solution

- ① Unique solution
- ② Infinitely many solution

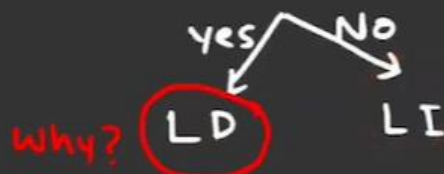
↑ Trivial solution is zero vector

Problem: If $Ax = 0$ has some solution, then what can you say about Linear dependency of columns of A?

→ let $A = [a_1 \ a_2 \ a_3]$ and $x = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$

$$c_1 a_1 + c_2 a_2 + c_3 a_3 = 0 \text{ where } c_1, c_2 \text{ and } c_3 \text{ are scalars}$$

Is there any non-zero c_i



Then any one vector can be expressed as a Linear combination of other vectors. Let's assume $c_3 \neq 0$

$$c_3 a_3 = -c_1 a_1 - c_2 a_2$$

$$a_3 = -\frac{c_1}{c_3} a_1 - \frac{c_2}{c_3} a_2 \leftarrow \text{here } a_3 \text{ is a linearly dependent vector}$$

Problem: Coefficient matrix $A_{3 \times 3}$ in $Ax=b$ contains 3 Linearly independent Columns. What can you infer about solution ($b \neq 0$)?

$$\rightarrow \text{let } A = [a_1 \ a_2 \ a_3] \text{ and } x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \text{ and } b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$x_1 a_1 + x_2 a_2 + x_3 a_3 = b$$

Any 3 LI vector in $\mathbb{R}^3$ will fill the entire space in $\mathbb{R}^3$

So solution always exists.

Problem: Coefficient matrix $A_{3 \times 4}$ in $Ax=b$ contains 3 Linearly independent Columns. What can you infer about solution ($b \neq 0$)?

$$\rightarrow \text{let } A = \begin{bmatrix} \uparrow & \uparrow & \uparrow & \uparrow \\ a_1 & a_2 & a_3 & a_4 \\ \downarrow & \downarrow & \downarrow & \downarrow \end{bmatrix} \text{ and } x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \text{ and } b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in \mathbb{R}^3$$

$$\text{each } a_i \in \mathbb{R}^3$$

$$x_1 a_1 + x_2 a_2 + x_3 a_3 + x_4 a_4 = b$$

$\hookrightarrow$ Any 3 L.I vector in $\mathbb{R}^3$ fills the entire space in $\mathbb{R}^3$

so solution always exists

Problem: Coefficient matrix $A_{5 \times 4}$ in $Ax=b$ contains 3 Linearly independent Columns. What can you infer about solution ($b \neq 0$)?

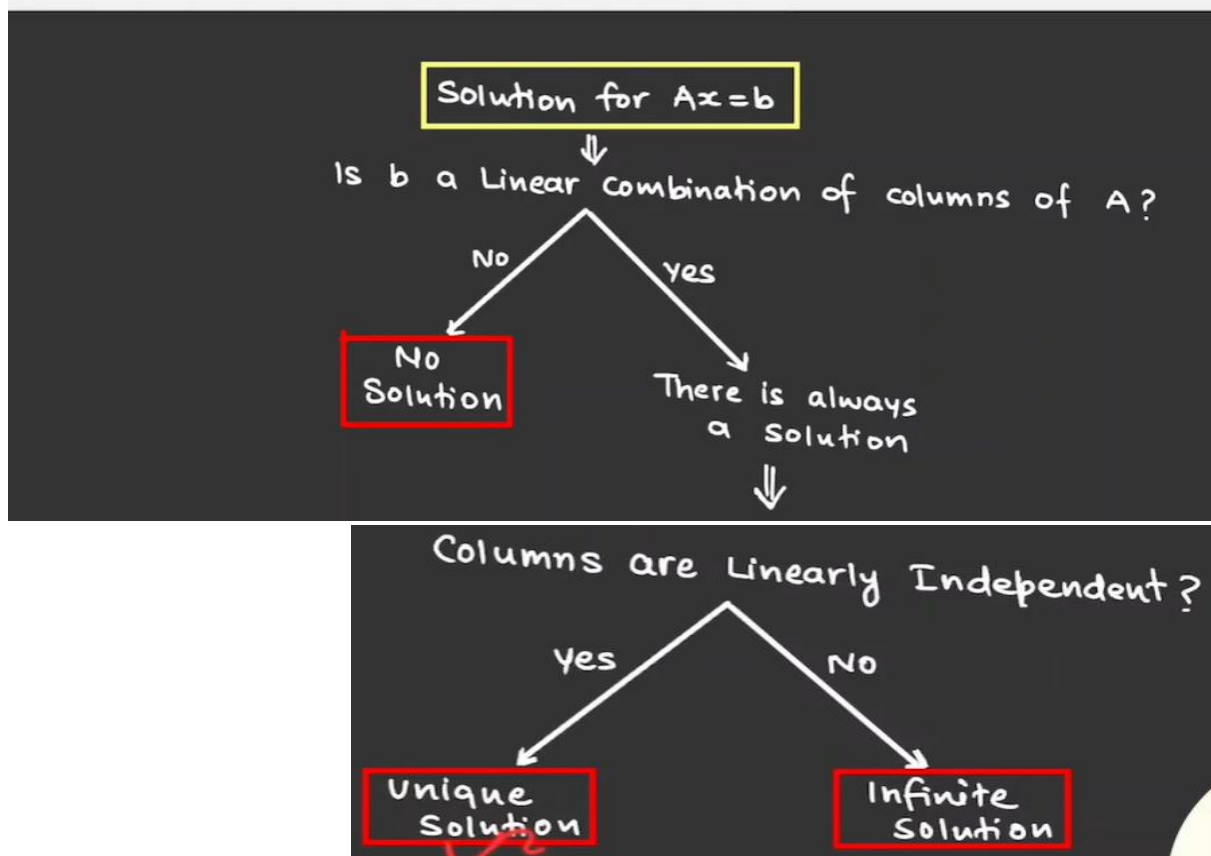
→ let $A = \begin{bmatrix} \uparrow & \uparrow & \uparrow & \uparrow \\ a_1 & a_2 & a_3 & a_4 \\ \downarrow & \downarrow & \downarrow & \downarrow \end{bmatrix}$ and $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix} \in \mathbb{R}^5$
each $a_i \in \mathbb{R}^5$

$$x_1 a_1 + x_2 a_2 + x_3 a_3 + x_4 a_4 = b$$

↳ Any 3 L.I vector in $\mathbb{R}^5$ can't fill the entire space in $\mathbb{R}^5$
so solution may or may not exist.

→ Till now, we saw whether solution exists or not for $Ax=0$

→ can we say whether unique/infinite solution?



Example: $A_{m \times n}$ ($m < n$) has m linearly independent columns, then there is always a solution of $Ax=b$?

no of equations $<$ no of variables

$$\rightarrow A_{m \times n} x_{n \times 1} = b_{m \times 1}$$

Here $b \in \mathbb{R}^m$

$$A = \begin{bmatrix} \uparrow & \uparrow & \uparrow & \dots & \uparrow \\ a_1 & a_2 & a_3 & \dots & a_n \\ \downarrow & \downarrow & \downarrow & \dots & \downarrow \end{bmatrix}$$

Each $a_i \in \mathbb{R}^m$

$\rightarrow$ we have m LI vectors and each vector $a_i \in \mathbb{R}^m$

$\rightarrow$ Any m LI vectors in $\mathbb{R}^m$ fills entire space in $\mathbb{R}^m$

$\rightarrow$ so there is always a solution $\leftarrow$ True